

UNIT-3

ELECTRO CHEMISTRY

Introduction:-

The branch of chemistry which deals with the study of relationship between electrical and chemical energy and their interconversion is called electro-chemistry."

- The basis of such processes is redox reactions. Redox reactions take place in various chemical and biological reactions.

Redox reactions are used in - burning of fuels, industrial processes, digestion of food by organisms, photosynthesis for getting energy from sun, metal extraction, preparation of important chemicals, preparation of dry & moist batteries, fuel cell etc.

- The chemical reactions which are take place by electrochemical processes are energy rich and produce less pollution.

Hence, the study of electrochemistry is very important to develop new eco-friendly techniques.

Biological system like transfer of sense actions from brain to cells to opposite to this is also coordinated by electrochemistry.

Metallic and Electrolytic Conductor:-

- * Conductor:- Those substances from which electric current can pass through are called conductors.

Metals are good conductor of electricity, e.g. Copper, silver, tin, etc.

Conductors can be classified into **two types -**

- (i) Metallic conductors:- Those conductors in which electricity is flow due to the presence of free e^- are called metallic conductor

→ The conductivity of metallic conductor depends upon the following factors:-

- (A) The nature and structure of the matter.
- (B) The no. of valence electron per atom.
- (C) The density of matter.
- (D) Temperature → If we increase the temperature then the conductivity of metallic conductor decrease.
(Temp ↑ → Conductivity ↓)

(ii) Electrolyte conductor:- Those conductor which do not flow electricity in solid-state but in molten state.

This type of conductor, conduct electricity due to the presence of free ions, are called electrolyte conductor.

→ Difference b/w Metallic and Electrolyte Conductor:-

Metallic Conductor	Electrolytic Conductor
1. In these conductors, conductivity is produced due to movement of e^- .	1. In these conductor conductivity is produced due to movement of free ions.
2. In metallic conductor, no chemical change take place during the conduction of electricity.	2. In these conductors, chemical change take place.
3. In metallic conductor flow of substance does not take place.	3. In these conductors flow of substance take place.
4. The conductivity of these conductor decrease with increase in temperature.	4. The conductivity of electrolyte increase with increase in temperature.

Classification of electrolytes → All electrolyte do not dissociate in same amount at all the given dilution.

On the basis, electrolytes are divided mainly in two parts - strong electrolytes and weak electrolytes.

(i) Strong electrolytes → "Strong electrolytes" are those substance which are completely ionized in their aqueous solution.

Generally, this class contains strong acids, strong bases and salt of strong acid-base (strong), salts of strong acid-weak base and weak acid-strong base.

e.g. HCl , NaOH , NaCl , CH_3COONa , NH_4Cl etc.

These salts are completely dissociated at normal dilution.

(ii) Weak electrolytes → "Weak electrolytes" are those substances which are not completely dissociated in their aqueous solutions and degree of dissociation increases with increase in solution.

Ionic part of weak electrolytes is called 'degree of dissociation' or 'degree of ionization' at given dilution. It is denoted by α .

This class contains weak acids, weak bases and salt of weak acid & weak base e.g. CH_3COOH , H_2CO_3 , H_3BO_3 , NH_4OH , $\text{Al}(\text{OH})_3$, $\text{CH}_3\text{COONH}_4$ etc.

→ Factors affecting the conductance of electrolyte →

The Conductance

of electrolyte depends upon the following factor.

(i) Inter-ionic attraction → If strong inter ionic attraction is present b/w the ions of solute then do not dissociate easily, when dissolve in solvent, due to this the value of conductivity become less.

→ The energy released in ion-solvent reaction is called 'solvation energy' and if water is solvent then it is called 'hydration energy'.

So, if the amount of ion-ion attraction energy is more than ion-solvent attraction then electrolyte does not dissociate.

This type of electrolytes are taken in class of weak electrolytes.

(ii) Solvation of ions: - It depends on attractive forces present between ions and solvents, which are called "ion-solvent interactions".

If this attractive force is very strong then ion become solvate in which the layers of solvent molecules are attached to ions and increase its size.

In this condition, the movement of ions towards electrode decreases, so the conductivity decreases.

(iii) Viscosity of solvent: - Viscosity of solvent depends on solvent-solvent interactions.

Greater the attraction, greater is the viscosity of solvent, which decreases the conductivity.

(iv) Concentration of solution: - Greater the conc. of electrolyte solution, lesser is the value of its conductivity.

The amount of dissociation increases with increase in dilution. Due to which, the conductivity of solution of electrolyte increases.

(v) Effect of temperature: - The average kinetic energy of ions of electrolyte increases with increase in temperature, which increases the degree of dissociation of electrolyte. Hence, amount of conductivity increases with increase in temp.

Cell $\rightarrow$ The device which convert electrical energy into chemical energy or chemical energy into electrical energy is called cell.

$\rightarrow$ cell are of the 2 types -

1. Electrochemical cell $\rightarrow$ The cell which convert chemical energy into electrical energy is called electrochemical cell.

Example $\rightarrow$ Daniel cell.

2. Electrolytic cell $\rightarrow$ The cell which convert electrical energy into chemical energy is called electrolytic cell.

Example $\rightarrow$ Electrolysis of NaCl

$\rightarrow$ In electrolytic cell redox reaction is take place.

$\rightarrow$ In electrolytic cell non-spontaneous ($\Delta G = +ve$) reaction type is proceed.

For this cell the value of ΔG is positive.

* Structure of electrolytic cell $\rightarrow$ In this cell both cathode and anode are taken in a same container and both electrode are connected to a battery and in container electrolyte is taken in molten state.

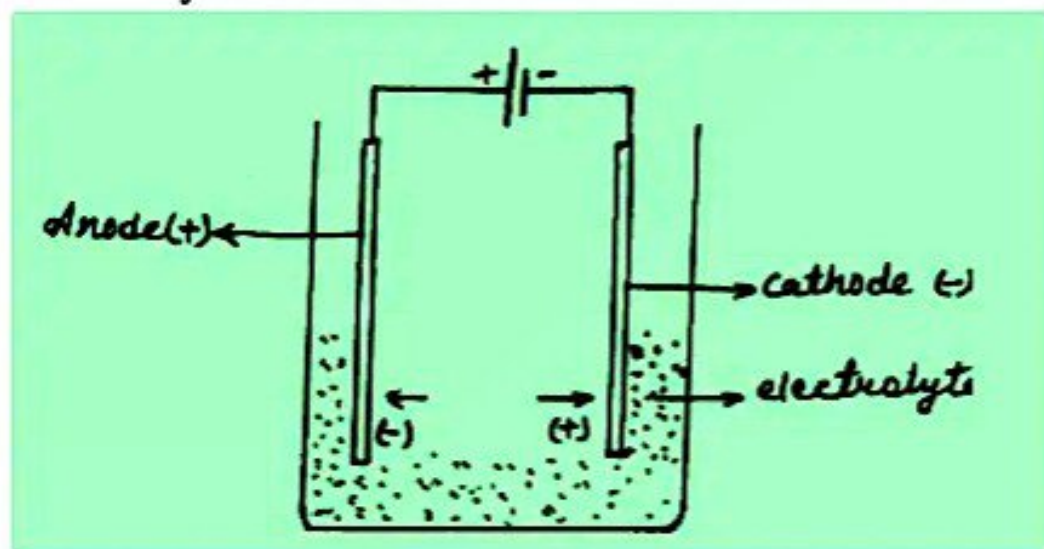
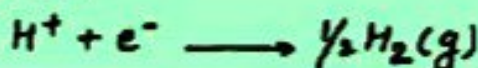


Fig $\rightarrow$ Structure of electrolytic cell.

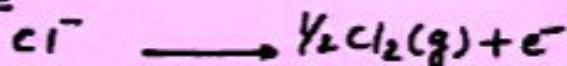
→ If HCl solution is take place in electrolytic cell then the following reaction take place at cathode and at anode.



At Cathode -



At anode -



→ Laws of Electrolysis → Michael Faraday give the law of electro-lysis.

[1.] Faraday first law of Electrolysis → According to this law, "The amount of electrolyte depo-sited at electrode is directly proportional to the amount of electricity passed out through the solution."

If the amount of electrolyte is 'w' and amount of electricity is 'Q'. Then-

$$w \propto Q$$

$$(Q = I \times t)$$

or

$$w = Z \cdot Q$$

or

$$\boxed{w = Z \cdot I \cdot t} \quad (Z = \text{electrochemical equivalent})$$

→ Electrochemical equivalent :-

$$w = Z \cdot I \cdot t$$

— (i) (from faraday's 1st law)

if $I = 1\text{A}$, $t = 1\text{sec}$, then

$$\boxed{w = Z}$$

→ "If 1 amp. electric current is passed out, through a electrolytic, solution for 1sec. then, the amount of electrolytic deposited at electrode is called electrochemical equivalent."

→ Unit of 'Z' →

$$w = Z \cdot I \cdot t$$

$$Z = \frac{w}{I \cdot t}$$

$$Z = \frac{\text{gm}}{\text{A} \cdot \text{sec.}}$$

$$\boxed{Z = \text{gmA}^{-1}\text{sec}^{-1}}$$

→ To calculate the amount of electrolyte deposited at an electrode. The following formula is used -

$$W = \frac{m \times I \times t}{n \times F}$$

here - m = atomic mass of electrolytic ion

n = valency of ion

F = 1 Faraday = 96500 C

I = Electric current

t = time (sec.)

Ques:- The solution of CuSO_4 is electrolysed by current of 1.5 A till 10 minutes, then what will be the mass of Copper deposited at cathode? ($F = 96500 \text{ C}$)

Ans -

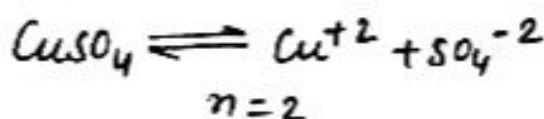
$$W = \frac{m \times I \times t}{n \times F}$$

Given - $I = 1.5 \text{ A}$

$t = 10 \text{ min}$

$t = 10 \times 60 = 600 \text{ sec.}$

$m(\text{Cu}) = 63.5 \text{ g mol}^{-1}$



$$W = \frac{63.5 \times 1.5 \times 600}{2 \times 96500}$$

$$W = 0.29 \text{ g}$$

[2] Faraday Second Law:- According to this law -
 "When the same amount of electricity is passed through different electrolyte then the amount of electrolyte deposited at electrode are in the ratio of their equivalent weight".

→ If the deposited amount of two electrolyte is respectively W_1 and W_2 and their equivalent weight is E_1 and E_2 then, according to this law -

$$\frac{W_1}{W_2} = \frac{E_1}{E_2}$$

— (1)

from 1st law of Faraday -

$$W_1 = Z_1 \cdot I \cdot T$$

$$W_2 = Z_2 \cdot I \cdot T$$

then from eqⁿ (1)

$$\Rightarrow \frac{Z_1 \cdot I \cdot T}{Z_2 \cdot I \cdot T} = \frac{E_1}{E_2}$$

$$\Rightarrow \frac{Z_1}{Z_2} = \frac{E_1}{E_2}$$

OR $Z \propto E$

→ Therefore the electrochemical equivalent of an electrolyte is directly proportional to the equivalent weight of electrolyte.

÷ 1-Faraday → The amount of electricity which is required to deposit 1 gm. equivalent weight of a substance at electrode is called 1 Faraday.

1 Faraday charge is equivalent to the charge present on 1 mole e^- .

$$\text{Charge on one } e^- = 1.6 \times 10^{-19} \text{ C}$$

$$\text{Charge on 1 mole of } e^- = 1.6 \times 10^{-19} \times 6.023 \times 10^{23} \\ = 96500 \text{ C}$$

$$\{ 1 \text{ Faraday} = 96500 \text{ C} \}$$

* Conductance in electrolytic solution →

÷ Ohm's Law → According to this law, "The electric current passed through a conductor is equal to the ratio of voltage and resistance of conductor."

$$I = \frac{V}{R}$$

÷ Resistance (R) → The obstruction in the flow of electric current in a conductor is called resistance.
→ It is represented by 'R'.

Unit of 'R' = Ohm (Ω)

÷ Conductance :- The reciprocal of resistance is called Conductance. It is represented by 'C'.

$$C = \frac{1}{R}$$

Unit of 'C' → Ohm⁻¹ or mho or Siemens (S)

÷ Specific resistance or resistivity :- The resistance (R) of a conductor is directly proportional to the length (l) of a conductor and inversely proportional to the area of cross section (A).

Then, $R \propto \frac{l}{A}$

$$R = \rho \frac{l}{A}$$

here → R ⇒ Resistance

l ⇒ length of a conductor

A ⇒ cross section area

ρ ⇒ Resistivity or specific resistance

→ If $l = 1\text{ cm}$ and $A = 1\text{ cm}^2$ then -

$$R = \rho$$

→ If the length of a conductor is 1 cm and area of cross section is 1 cm², then the resistance produced by the conductor is called specific resistance or Resistivity.

unit of ' ρ ' ⇒

$$R = \rho \cdot \frac{l}{A}$$

$$\rho = R \cdot \frac{A}{l}$$

$$\Rightarrow \rho = \text{ohm} \cdot \frac{\text{cm}^2}{\text{cm}}$$

$$\Rightarrow \boxed{\rho = \text{ohm} \cdot \text{cm}}$$

Specific Conductance or Conductivity: $\rightarrow$ The reciprocal of specific resistance is called specific conductance or conductivity.

$\rightarrow$ It is represented by ' κ '. (Kappa)

$$\boxed{\kappa = \frac{1}{\rho}}$$

$$\kappa = \frac{1}{R} \times \frac{l}{A} \rightarrow \text{Unit of } \kappa = \frac{1}{\text{ohm}} \cdot \frac{\text{cm}}{\text{cm}^2}$$

$$\boxed{\kappa = C \cdot \frac{l}{A}}$$

$$\boxed{\kappa = \text{ohm}^{-1} \text{ cm}^{-1}}$$

Specific conductance -

$$\kappa = C \times \frac{l}{A}$$

if $l = 1 \text{ cm}$, $A = 1 \text{ cm}^2$ then -

$$\boxed{\kappa = C}$$

$\rightarrow$ If the length of conductor is 1 cm and area of cross section is 1 cm^2 then the conductance produced by the conductor is called specific conductance.

Cell constant (α) $\rightarrow$ The ratio of the length of conductor (l) and area of cross section (A) is called cell constant.

$$\boxed{\alpha = \frac{l}{A}} \Rightarrow \left\{ \frac{\text{cm}}{\text{cm}^2} = \text{cm}^{-1} \right\} \rightarrow \text{unit of } \alpha$$

* Relation between conductance, specific conductance and cell-constant is given by the following equation -

$$\text{Specific conductance} = \text{Conductance} \times \text{cell constant}$$

$$\{ \kappa = c \cdot x \}$$

† Molar Conductance $\rightarrow (\lambda_m) \rightarrow$ If 1 mole of an electrolyte is dissolved in 1000 ml & placed between two electrodes having 1 cm. distance and 1 cm² area of cross section then the conductance produced by the ion is called molar conductance.

$$\lambda_m = \frac{\kappa \times 1000}{c} \quad c = \frac{\text{Mole}}{\text{litre}}$$

unit of λ_m :-

$$\lambda_m = \frac{\kappa \times 1000}{c} = \frac{\text{ohm}^{-1} \text{cm}^{-1} \times \text{cm}^3}{\text{mole}}$$

$$\lambda_m = \text{ohm}^{-1} \text{cm}^2 \text{mol}^{-1}$$

† Equivalent Conductance $\lambda_v \rightarrow$ When 1 gm. equivalent of an electrolyte is dissolved in 1000 ml and placed between two electrodes having 1 cm. distance and 1 cm. area of cross section then the conductance produced by the ion is called equivalent conductance.

$$\lambda_v = \frac{\kappa \times 1000}{c} \quad (c = \text{equivalent weight})$$

unit of λ_v :- $\lambda_v = \frac{\text{ohm}^{-1} \text{cm}^{-1} \times \text{cm}^3}{\text{gram equivalent}}$

$$\lambda_v = \text{ohm}^{-1} \text{cm}^2 \text{g eq}^{-1}$$

→ Effect of dilution on conductance, conductivity and molar conductivity :-

If the dilution of an electrolytic solution is take place then the no. of mole of electrolyte is increased with increase in volume of solution. then the molar conductance is increased with dilution.

→ The specific conductance of an electrolyte solution is depends upon the no. of mole of ion present in a specific area but if we increase the dilution then the total no. of ion is increased but in specific area no. of mole of ions is decrease due to this specific conductance decrease with increase in dilution.

Kohlrausch law :- Kohlrausch draw a graph between molar conductivity of various electrolyte and square root of the conc. for strong and weak electrolyte on the basis of above equation.

$$\Lambda_m = \Lambda^\infty - b\sqrt{c}$$

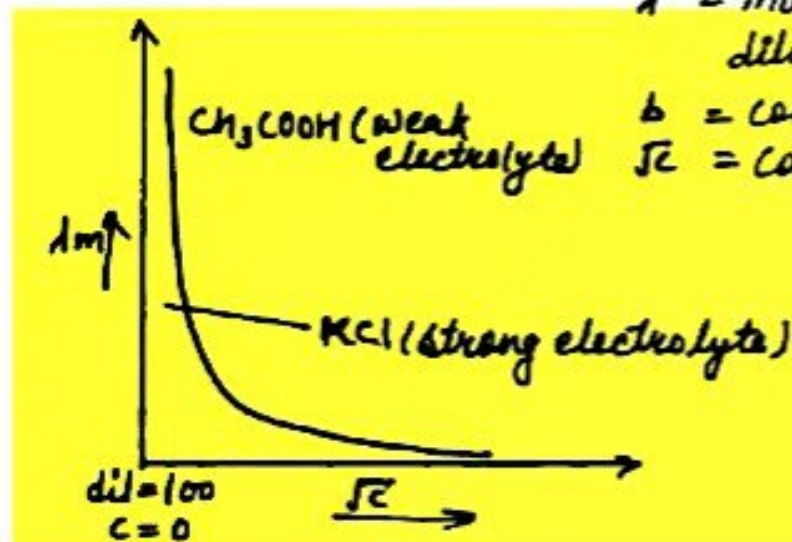
here -

Λ_m = molar conductance

Λ^∞ = molar conductance at infinite dilution.

b = constant

$\sqrt{c}$ = Concentration



→ The following conclusions are obtained from above graph -

- strong electrolyte give straight line and weak electrolyte give non-linear line.
- The molar conductance of weak electrolyte is continuous increase

with increase in dilution or decrease in concentration but the molar conductance of strong electrolyte is increase at a certain dilution and after this it become constant.

(c) The molar conductance of an electrolyte at 100% dilution is called molar conductance at infinite dilution.

→ Kohlrausch's law of independent migration of ions →

According to this law -
"The molar conductance of an electrolyte at infinite dilution is equal to the sum of molar conductance of electrolyte ion at infinite dilution present in electrolyte."

$$\lambda_m^\infty = \lambda_+^\infty + \lambda_-^\infty$$

OR

$$\lambda_m^\infty = \nu_+ \lambda_+^\infty + \nu_- \lambda_-^\infty$$

here - ν_+ and ν_- = No. of cation and anion

E.g: $\lambda_{NaCl}^\infty = \lambda_{Na^+}^\infty + \lambda_{Cl^-}^\infty$

$$\lambda_{CH_3COOH}^\infty = \lambda_{CH_3COO^-}^\infty + \lambda_{H^+}^\infty$$

$$\lambda_{MgCl_2}^\infty = \lambda_{Mg^{+2}}^\infty + 2\lambda_{Cl^-}^\infty$$

$$\lambda_{AlCl_3}^\infty = \lambda_{Al^{+3}}^\infty + 3\lambda_{Cl^-}^\infty$$

⇒ Application of Kohlrausch Law :-

(a) Calculation of molar conductance of electrolyte at infinite dilution :-

The molar conductance of a weak electrolyte is calculate by the help of the molar conductance of strong electrolyte.

Ex. CH_3COOH is a weak electrolyte then the molar conductance of this electrolyte at infinite dilution is calculate by help of strong electrolyte CH_3COONa , NaCl and HCl .

Ans. $\lambda_{\text{CH}_3\text{COOH}}^{\infty} = \lambda_{\text{CH}_3\text{COO}^-}^{\infty} + \lambda_{\text{H}^+}^{\infty}$

$$\lambda_{\text{CH}_3\text{COONa}}^{\infty} = \lambda_{\text{CH}_3\text{COO}^-}^{\infty} + \lambda_{\text{Na}^+}^{\infty} \quad \text{--- (1)}$$

$$\lambda_{\text{NaCl}}^{\infty} = \lambda_{\text{Na}^+}^{\infty} + \lambda_{\text{Cl}^-}^{\infty} \quad \text{--- (2)}$$

$$\lambda_{\text{HCl}}^{\infty} = \lambda_{\text{H}^+}^{\infty} + \lambda_{\text{Cl}^-}^{\infty} \quad \text{--- (3)}$$

from eq. (1) + eq (3) - eq (2)

$$\lambda_{\text{CH}_3\text{COONa}}^{\infty} + \lambda_{\text{HCl}}^{\infty} - \lambda_{\text{NaCl}}^{\infty} = \lambda_{\text{CH}_3\text{COO}^-}^{\infty} + \cancel{\lambda_{\text{Na}^+}^{\infty}} + \lambda_{\text{H}^+}^{\infty} + \cancel{\lambda_{\text{Cl}^-}^{\infty}} - \cancel{\lambda_{\text{Na}^+}^{\infty}} - \cancel{\lambda_{\text{Cl}^-}^{\infty}}$$

$$\lambda_{\text{CH}_3\text{COONa}}^{\infty} + \lambda_{\text{HCl}}^{\infty} - \lambda_{\text{NaCl}}^{\infty} = \lambda_{\text{CH}_3\text{COO}^-}^{\infty} + \lambda_{\text{H}^+}^{\infty}$$

$$\lambda_{\text{CH}_3\text{COONa}}^{\infty} + \lambda_{\text{HCl}}^{\infty} - \lambda_{\text{NaCl}}^{\infty} = \lambda_{\text{CH}_3\text{COOH}}^{\infty}$$

(b) Calculation of degree of dissociation of a weak electrolyte →

→ The amount of a weak electrolyte which is dissociate into ions is called degree of dissociation and it calculate by using the following formula.

$$\alpha = \frac{\text{Molar conductance at a given concentration}}{\text{Molar conductance at infinite solution}}$$

$$\alpha = \frac{\lambda_c}{\lambda_{\infty}}$$

(c) Transport number of ion :- The ratio of molar conductivity of electrolyte at infinite dilution

and molar conductivity of electrolyte at infinite dilution is called transport no. of ions.

$$\text{Transport no. of cation } (\eta_c) = \frac{\lambda_c^\infty}{\lambda^\infty}$$

$$\eta_c = \frac{\lambda_c^\infty}{\lambda^\infty}$$

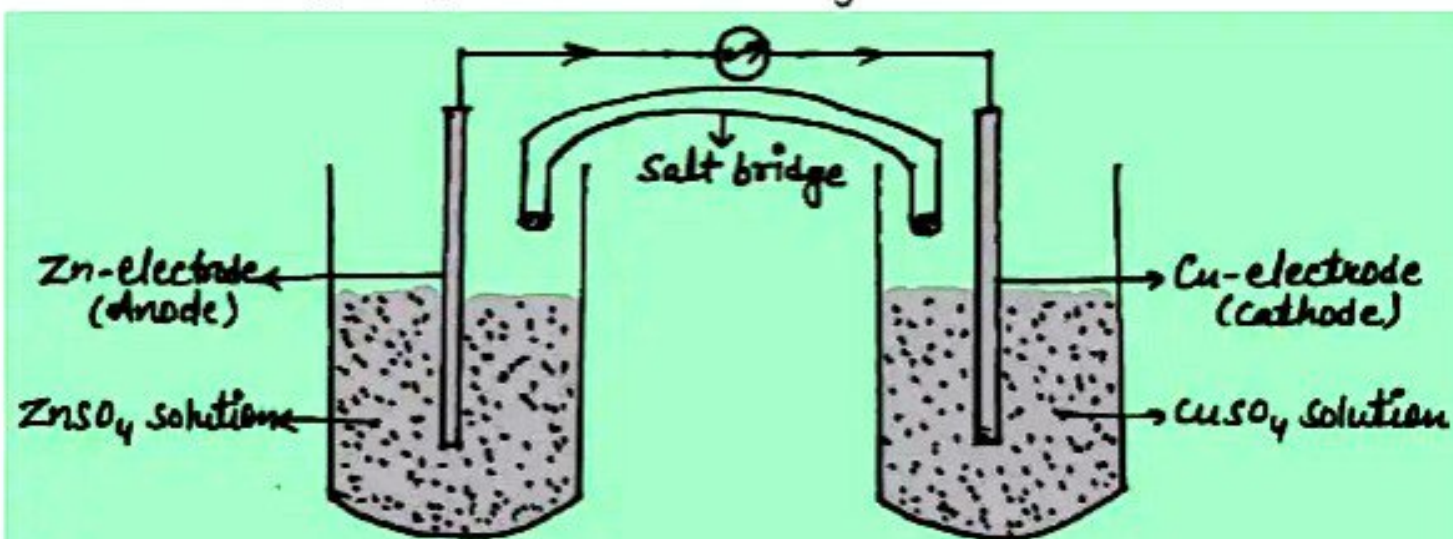
$$\text{transport no. of anion } \left\{ (\eta_a) = \frac{\lambda_a^\infty}{\lambda^\infty} \right\}$$

Electrochemical cell: → The cell in which chemical energy is converted into electrical energy is called electrochemical cell.

These cell was discovered by 'Lewis Galvanic' & 'Alexander Volta' then these are also called galvanic or voltaic cell.

→ In this cell redox reaction is take place which is spontaneous process.

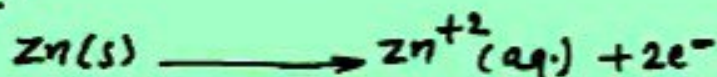
→ The value of ΔG for these cell is negative.



Galvanic or Daniel cell

→ Reaction in electrochemical cell :- When both electrode are joined together by a wire then the following reaction take place at cathode and at anode.

→ At anode -



→ At cathode -



Overall reaction:-



→ Electrochemical cell is represented by the following formula -

L → Left cell,
O → Oxidation
A → Anode
N → Negative

★ Salt bridge → The Inverse U-shaped tube which connects both half cells is called salt bridge.

It is a glass tube in which paste of some electrolyte like - KCl , Na_2SO_4 is filled up.

→ Function of salt bridge:-

1. Salt bridge connects both half cells and completes the circuit of the cell.
2. Salt bridge maintains the neutrality of both half cells.

⇒ Representation of Galvanic Cell by Symbol System →

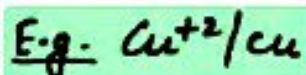
The following rules are used to represent a galvanic cell by symbol system.

1. To write the cell in symbol system first we write anode on the left hand side and then write cathode on the right hand side.
2. Both cathode and anode are separated by two vertical lines which represent salt bridge.

3. To write anode first we write metal atom and then metal ion and both are separate by vertical line.

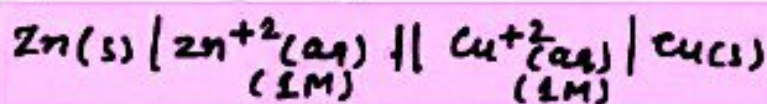


4. To write cathode first we write metal ion and then metal atom and both are separate by vertical line.



5. After the metal ion the concentration of electrolyte is represent in small bracket.

→ The symbol representation of Galvanic cell is take place as:



Electrode Potential: → The potential difference which is set-up between the metal electrode and its electrolytic solution is called Electrode potential.

→ It is represented by 'E'.

→ Electrode potential are of the two types -

(a) Oxidation potential

(b) Reduction potential

(a) Oxidation potential: → The electrode potential of oxidation half-cell is called Oxidation potential.

→ It is represent by 'E_{oxi}'.

(b) Reduction potential: → The electrode potential of reduction half-cell is called reduction potential.

→ It is represent by 'E_{red}'.

→ for a metal electrode the value of oxidation potential and reduction potential is equal, but there sign are oppside.

$$E_{\text{oxi}} = - E_{\text{red}}$$

**** Standard Electrode potential :-** At 298 K temperature, if the concentration of electrolyte is 1M then the electrode potential is called standard electrode potential.

→ It is represented by 'E'.

**** Cell potential or emf of cell** → In open circuit the difference of potential b/w two half cell is called cell potential or emf of cell.

→ The cell potential or emf of cell is calculated by using the following formula.

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}} \quad (\text{Only take reduction})$$

At Standard E.P. Condition -

$$\{ E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}} \}$$

⇒ **Potential difference** :- When circuit is closed when the potential difference between the electrode of two half cell is called potential difference.

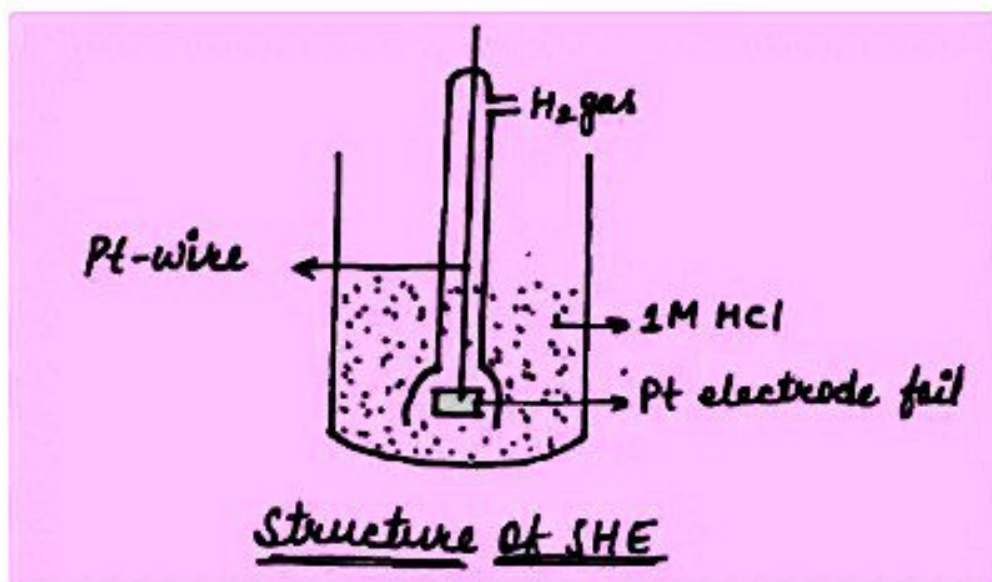
**** Difference b/w cell potential & potential difference** →

Cell potential	Potential Difference
1. It is potential difference b/w two electrode when circuit is open.	1. It is potential difference b/w two electrode when circuit is closed.
2. It is equal to maximum emf of the cell.	2. It is always less than maximum emf of the cell.
3. It is measured by potentiometer.	3. It is measured by Voltmeter.

Standard Hydrogen Electrode (SHE) → It is a reference electrode which is used to measure the electrode potential of a half cell.

→ The standard electrode potential for hydrogen electrode is zero (0).

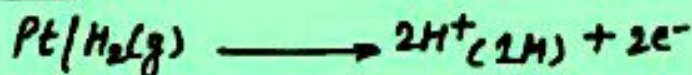
→ This cell can act as both oxidation or reduction cell.



→ In the structure of SHE a Pt (Platinum) wire is dipped in a glass tube which is filled by mercury (Hg) and at a lower end of Pt wire a rectangular Pt foil is present and in cell 1M concentration HCl solution is taken as a electrolyte and H₂ gas is passed at 1 atm. (atmospheric pressure) in glass tube.

→ SHE is represent as when act as cathode or anode.

As anode -



As cathode -



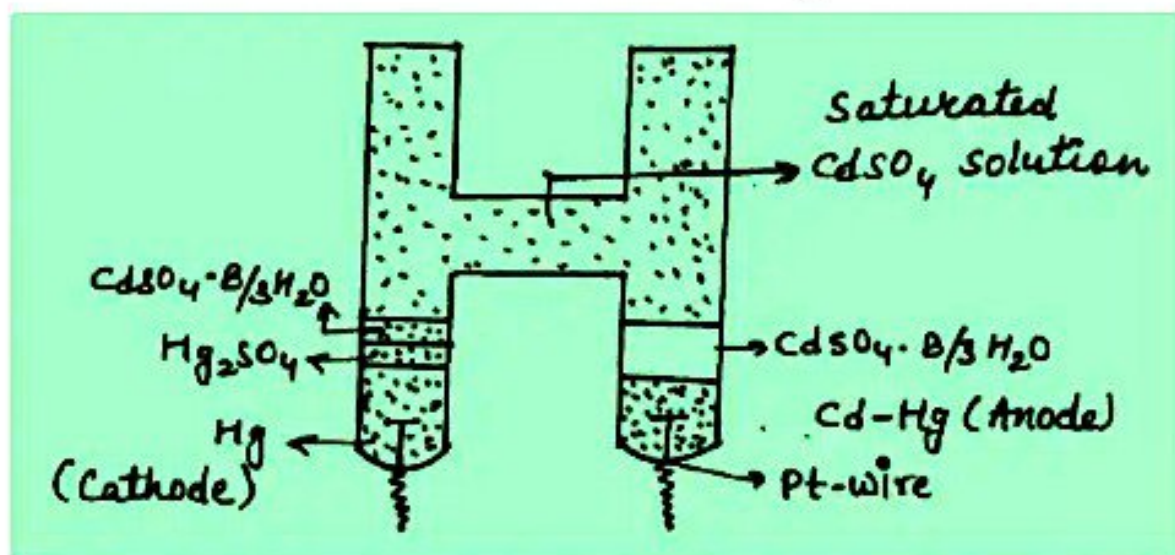
Standard Weston Cell :- This cell have some important property-

(i) It is a rigid cell which give accurate value of electro-motive force (EMF).

(ii) It is more reversible.

(iii) It is also called cadmium cell.

→ Structure of standard Weston cell:- It has H-shaped glass tube in which Pt-wire is fitted at lower end.

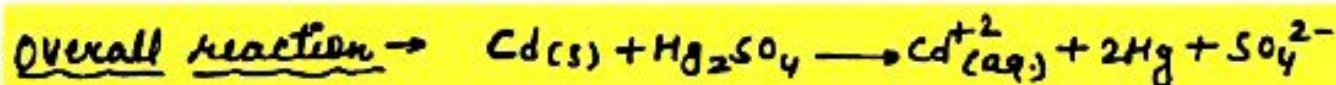
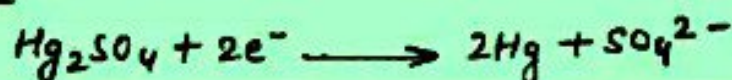


→ When Weston cell is used then the following reaction is take place at cathode and at anode -

At anode →



At cathode →



**** Electromotive force (emf) and Gibbs energy :-**

In electrochemical cell chemical energy is converted into electrical energy then the work which is done by cell is called electric work.

Electric work is equal to the product of amount of electric current and Emf of cell at constant temperature & pressure.

Then,

$$\text{Electric work (W)} = nFE$$

→ The electric work which is done by cell is equal to the decrease in free energy.

Then,

$$-\Delta G = nFE$$

$$\Delta G = -nFE$$

→ In standard condition-

$$\Delta G^\circ = -nFE^\circ$$

$$E = \frac{-\Delta G}{nF}$$

$$E \propto (-\Delta G)$$

→ Therefore the electromotive force of cell is directly proportional to the decrease in Gibbs free energy.

Nernst's Equation :-

Nernst derive a equation which represent the relation b/w Emf of cell, nature of electrode, temperature and the concentration of solution. Suppose the following reaction is take place in a reversible Galvanic cell:-



According to law of mass action -

$$K = \frac{[C]^c [D]^d}{[A]^a [B]^b} \quad \text{--- (i)}$$

→ According to thermodynamics, there is a following relation in Gibbs free energy and emf of cell.

$$\Delta G = \Delta G^\circ + RT \ln K \quad \text{--- (ii)}$$

→ There is a following relation in Gibbs free energy and emf of cell.

$$\Delta G = -nFE \quad \text{--- (iii)}$$

$$\Delta G^\circ = -nFE^\circ \quad \text{--- (iv)}$$

→ Put the value of ΔG and ΔG° in equation (ii) -

$$-nFE = -nFE^\circ + RT \ln K$$

$$nFE = nFE^\circ - RT \ln K \quad \text{--- (v)}$$

Divide equation (v) by $nF \rightarrow$

$$E = E^\circ - \frac{RT \ln K}{nF}$$

Put the value of 'K' from equation - (i) $\rightarrow$

$$E = E^\circ - \frac{2.303RT}{nF} \log \frac{[C]^c [D]^d}{[A]^a [B]^b} \quad \text{--- (vi)}$$

→ This is normal representation of Nernst's equation -
In equation (vi) if we put the value of R, T & F then equation (vi) can be written as:-

$$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$T = 298 \text{ K}$$

$$F = 96500 \text{ C}$$

$$E = E^\circ - \frac{0.0591}{n} \log \frac{[C]^c [D]^d}{[A]^a [B]^b} \quad \text{--- (vii)}$$

→ This equation (vii) is the platform of Nernst's equation.

Applications of Nernst's Equation → With the help of Nernst's equation, we can determine the cell potential of an electrode or half cell.

For Daniel cell Nernst's equation is written as:-



→ From Nernst's equation -

$$E = E^{\circ} - \frac{0.0591}{n} \log \frac{[P]}{[R]}$$

→ For this $n = 2$

$$E = E^{\circ} - \frac{0.0591}{2} \log \frac{[Zn^{+2}]}{[Cu^{+2}]}$$

→ The Concentration of $Zn(s)$ and $Cu(s)$ is not written in K_c because these are in solid state and these concentration are considered unity (1).

Ques 5 - Calculate the standard Electrode potential of Ni^{+2}/Ni electrode if the cell potential of cell potential of cell $Ni/Ni^{+2}_{(1M)} || Cu^{+2}_{(1M)}/Cu$ is 0.59 volt and given $E_{Cu^{+2}/Cu} = 0.34V$

Ans - The cell reaction is $Ni/Ni^{+2}_{(1M)} || Cu^{+2}_{(1M)}/Cu$

$$E^{\circ}_{cell} = E^{\circ}_{cathode} - E^{\circ}_{Anode}$$

$$E^{\circ}_{cell} = E^{\circ}_{Cu^{+2}/Cu} - E^{\circ}_{Ni^{+2}/Ni}$$

Given -

$$E^{\circ}_{cell} = 0.59V$$

$$E^{\circ}_{Cu^{+2}/Cu} = 0.34V$$

put the value of E°_{cell} & $E^{\circ}_{Cu^{+2}/Cu}$ in above equation -

$$0.59 = 0.34 - E^{\circ}_{Ni^{+2}/Ni}$$

$$\begin{aligned} E^{\circ}_{Ni^{+2}/Ni} &= 0.34 - 0.59 \\ &= -0.25V \end{aligned}$$

Ques:- If the standard electrode potential of copper electrode is $+0.34V$. Calculate electrode potential of $0.1M$ electrolyte containing copper ions (Given $\log 10 = 1$)

Ans:- At copper electrode the following reaction is take place at cathode:



From Nernst's equation -

$$E = E^{\circ} - \frac{0.0591}{n} \log \frac{[P]}{[R]}$$

Given:-

$$E^{\circ} = +0.34V$$

$$n = 2$$

$$E = ?$$

$$E = 0.34 - \frac{0.0591}{2} \log \frac{1}{[Cu^{+2}]}$$

$$\therefore [Cu^{+2}] = 0.1$$

$$E = 0.34 - \frac{0.0591}{2} \log \frac{1}{0.1}$$

$$E = 0.34 - \frac{0.0591}{2} \times 1$$

$$E = 0.3104V \approx 0.31V$$

Batteries or Cells:- When two or more galvanic cell are connect in series order then these are called Batteries or cells.

In a battery or cell the following property should be present-

1. These should be light weighted and compact.

2. These should produce constant voltage for a long time.

→ Batteries or cell are of the two types-

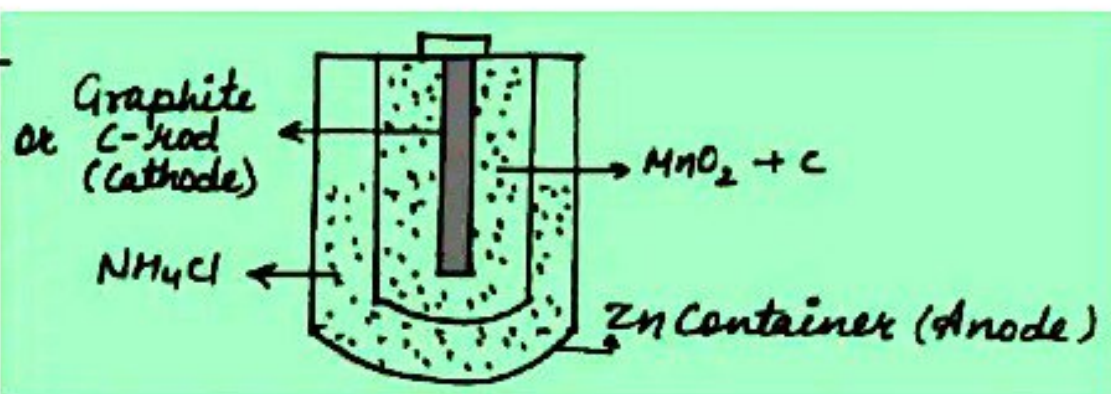
(A) **Primary cell or Battery:-** Those cell which cannot be recharged after the use are called primary cell or Battery.

In these cell the reaction which is take place is irreversible in nature.

Eg. Dry cell, Mercury cell.

(i) **Dry cell or Leclanche cell:-** These cell are developed by Leclanche then these are also called Leclanche cell.

Structure -

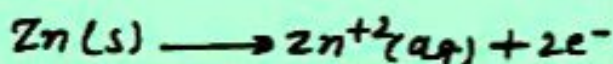


→ In this cell Zn container act as anode and graphite and C-rod (Carbon rod) act as cathode.

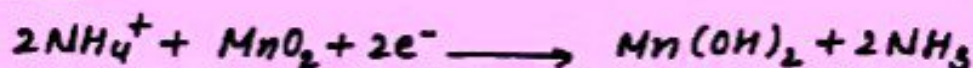
Around the cathode a paste of $MnO_2 + C$ is filled up and NH_4Cl act as a electrolyte.

→ **Mechanism:-** When dry cell is used then the following reaction is take place at cathode or anode.

At anode -



At Cathode -



Overall reaction -



→ In overall reaction the Zn^{+2} which is formed react with ammonia (NH_3) and form of complex ion $[\text{Zn}(\text{NH}_3)_4]^{2+}$.

→ The cell potential of dry cell is 1.2 to 1.5 Volt.

(ii) Mercury Cell :→ This cell is used in low voltage device like watch, hearing device, calculator etc.

→ In mercury cell Zn-Hg amalgam act as anode and paste of HgO act as Cathode and KOH is used as a electrolyte.

→ When mercury cell is used then the following reaction is take place at Cathode and anode.

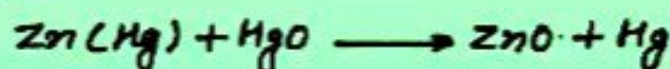
At anode -



At Cathode -



Overall reaction -



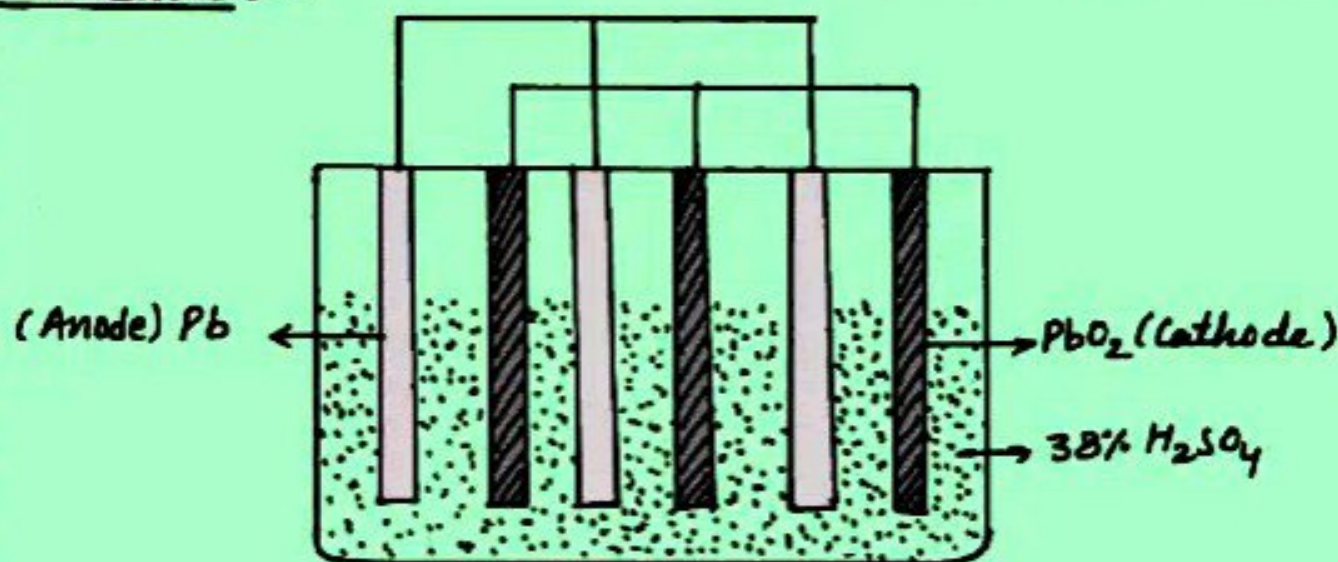
→ The cell potential of mercury cell is 1.35 Volt.

(B) Secondary cell or Battery:- The cell which can be recharged after the use are called secondary cell. In this cell the chemical reaction which is take place is reversible in nature.

E.g. Lead storage cell, Ni-cd cell.

(i) Lead storage cell:- It is a secondary cell and in this cell anode is made up of Pb and cathode is made up of PbO₂ and 38% H₂SO₄ solution act as electrolyte.

Structure:-



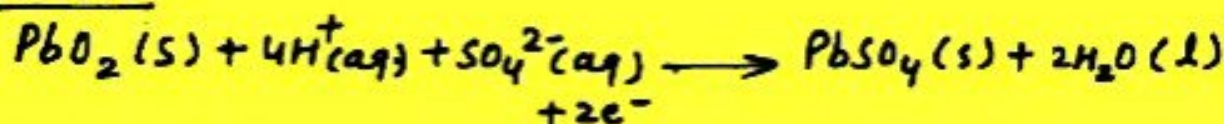
Lead Storage Cell

→ When the Lead Storage cell is used then the following reaction is take place at Cathode and anode.

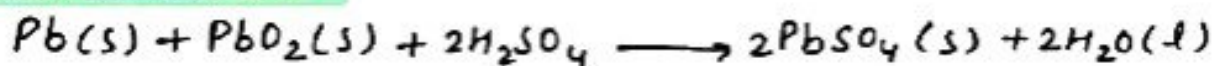
At anode -



At Cathode -



Overall Reaction :-



→ In overall reaction Pb and PbO₂ is convert into PbSO₄ and H₂SO₄ is convert into water then its concentration is decrease.

→ When lead storage cell is charged then during the charging the following reaction is take place.



→ The cell potential of lead storage, cell is 6 to 12 V (Volt).

(ii) Nickel-Cadmium :- It is a secondary cell and its work duration and efficiency is more than compare to lead-storage cell.

→ In this cell Cadmium (Cd) act as anode and NiO₂ grid act as Cathode, and KOH is used as electrolyte.

→ When Ni-cd cell is used then the following reaction is take place at cathode and anode.

At anode →



At Cathode →



Overall reaction →

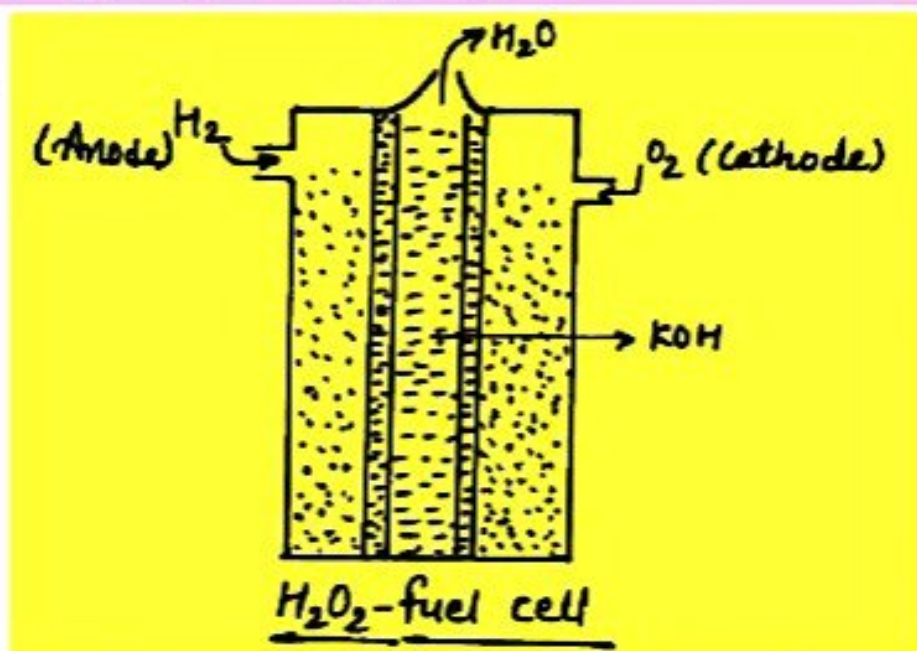


→ The cell potential of Ni-cd cell is 1.4 Volt.

***Fuel Cell:** → The galvanic cell in which combustion energy of fuel is directly convert into electrical energy is called fuel cell.

E.g. H_2O_2 -fuel cell (H_2 -Hydrogen, O_2 -Oxygen)
(Hydrogen-Oxygen fuel cell)

Structure →



→ In the structure of H_2O_2 fuel cell there are three compartments. In one compartment H_2 gas is passed which acts as anode. In another compartment O_2 gas is passed which acts as cathode and in the middle compartment KOH is used as an electrolyte.

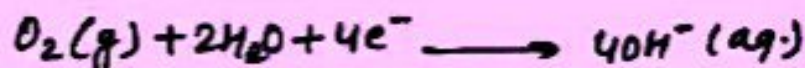
These compartments are separated by porous carbon electrodes in which finely divided Pt or Pd metal are filled as a catalyst.

***Mechanism:** → When fuel cell is used then the following reaction takes place at cathode and at anode.

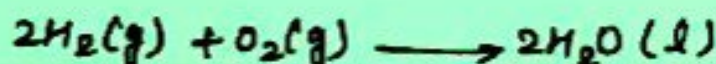
At anode →



At Cathode →



Net Reaction →



→ Characteristics of fuel cell :-

- (i) The capacity of fuel cell is high approximately $\approx 70\%$.
- (ii) These cell works continuously till the reactant are present
- (iii) These cell are pollution free.
- (iv) Fuel cell work for a long time.

**** Corrosion :-** The process in which active metal when comes in the contact of atmospheric gases and humidity then it is convert into its oxide, hydroxide or carbonate this process is called Corrosion.

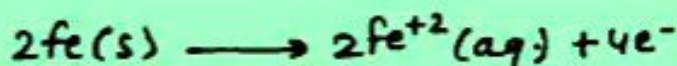
E.g.

- (i) Rusting of Iron
- (ii) Tarnishing of Silver
- (iii) Green colour substance deposited on copper.

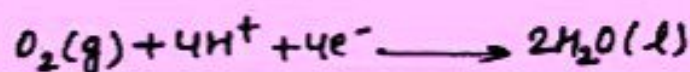
⇒ Rusting of Iron :- The process of rusting of iron is explained by electrochemical theory.

- In the process of rusting of iron a impure part of iron act as cathode, and the H^+ ion obtained from H_2CO_3 is act as electrolyte.
- H_2CO_3 is formed by dissolving of CO_2 in water.
- The following reaction is take place in the rusting of iron process.

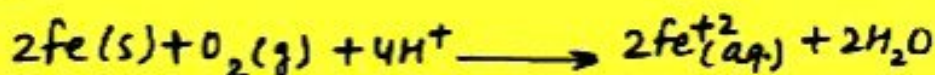
At anode -



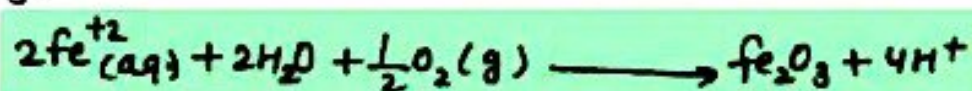
At cathode -



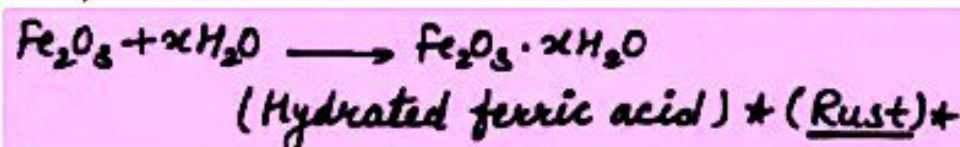
Net Rxn :-



→ In net reaction the 2Fe^{2+} ion which is form is float in water at iron surface then these are oxidised by atmospheric oxygen into Fe_2O_3 .



→ Now Fe_2O_3 combine with water molecule and the formation of rust is take place.



**Method of prevention of corrosion:-

(i) The most suitable method of corrosion is that prevent the metal from direct contact of atmosphere.

(ii) Corrosion can be prevent by coating the metal surface with suitable chemicals like - Oil paint.

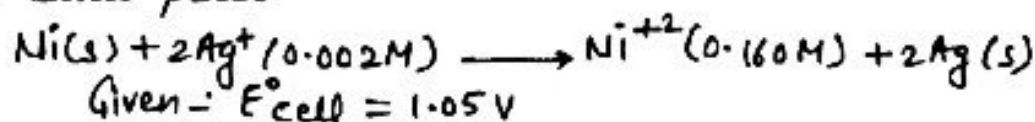
(iii) Sacrificing protection → The process in which less reactive metal is coated on the surface of more reactive metal, to protect its corrosion.

E.g. The process in zn is coated on the surface of fe. This process is called galvanisation.

Ques: Can you put the Copper sulphate solution in a zn pot?

Ans: Copper sulphate solution can not be stored in a zn pot because zn is more reactive than copper from copper sulphate solution, then we cannot store CuSO_4 in zn pot.

Ques: Calculate the emf of the cell in which the following reaction takes place -



Ans - From Nernst equation -

$$E = E^{\circ} - \frac{0.0591}{n} \log \frac{[P]}{[R]}$$

Given -

$$E^{\circ} = 1.05 \text{ V}, n = 2$$

$$[\text{Ni}^{+2}] = 0.160, [\text{Ag}^{+}]^2 = 0.002$$

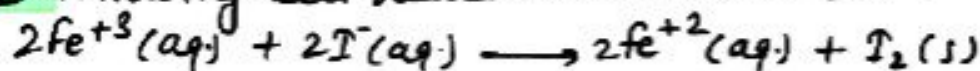
$$E = 1.05 - \frac{0.0591}{2} \log \frac{[\text{Ni}^{+2}]}{[\text{Ag}^{+}]^2}$$

$$E = 1.05 - \frac{0.0591}{2} \log \frac{[0.160]}{[0.002]^2}$$

$$E = 1.05 - \frac{0.0591}{2} \times 4.6021$$

$$E = 0.916 \text{ V}$$

Ques:- following cell reaction occurs in cell:



At 298 K the standard cell potential of this cell $E^{\circ}_{\text{cell}} = 0.236 \text{ V}$. Calculate standard Gibbs free energy and the equilibrium constant of the cell reaction.

Ans -

$$\Delta G^{\circ} = -nFE^{\circ}$$

Given $\rightarrow n = 2, F = 96500 \text{ C}, E^{\circ} = 0.236 \text{ V}$

$$\Delta G^{\circ} = -2 \times 96500 \times 0.236$$

$$= -45.55 \text{ kJ mol}^{-1}$$

$$\Delta_r G^{\circ} = -2.303 RT \log K_c$$

$$\Rightarrow \log K_c = \frac{-\Delta G^{\circ}}{2.303 RT}$$

$$\Rightarrow \log K_c = \frac{-45.55}{2.303 \times 8.314 \times 10^{-3} \times 298}$$

$$\log K_c = 7.983$$

$\Rightarrow$

$$\Rightarrow K_c = \text{antilog}(7.983) = 9.616 \times 10^7$$

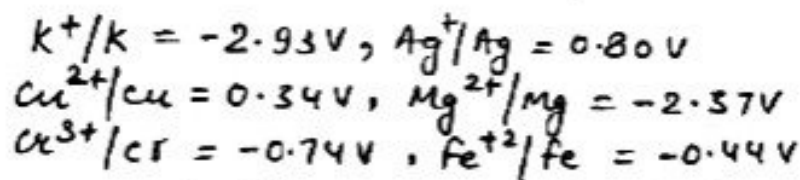
Question & Answers

Ques: 1. Define electrochemical cell. What happens if external potential applied becomes greater than E°_{cell} of electrochemical cell?
[All India 2016]

Ans. The device through which chemical energy changes to electrical energy is called electrochemical cell.

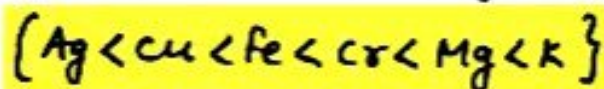
The chemical reaction of the cell is reversed and current flows in opposite direction when opposing emf is slightly greater than that of the cell.

Ques: 2. Given that, the standard electrode potentials (E°) of metals are -



Arrange these metals in an increasing order of their reducing power.
[All India 2010]

Ans. Metals in an increasing order of their reducing power -
 $\text{Ag}^+/\text{Ag}, \text{Cu}^{2+}/\text{Cu}, \text{Fe}^{2+}/\text{Fe}, \text{Cr}^{3+}/\text{Cr}, \text{Mg}^{2+}/\text{Mg}, \text{K}^+/\text{K}$



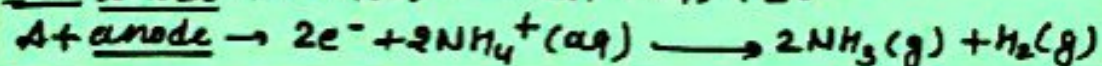
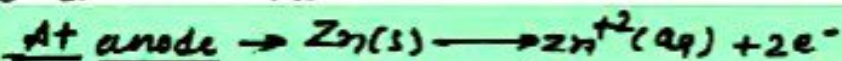
(more negative E° means that the redox couple is a stronger reducing agent)

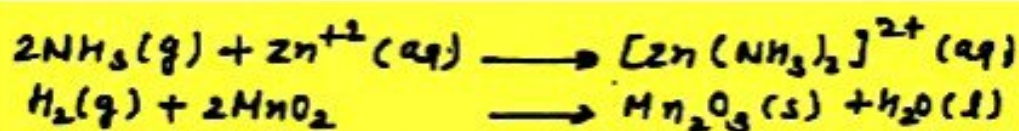
Ques: 3. Set-up Nernst's equation for the standard dry cell. Using this equation show that the voltage of a dry cell decreases with use.
[All India 2014C]

Ans. Dry cell (Leclanche cell) consists of a Zn container which acts as anode and Cathode is a Carbon (graphite) rod surrounded by powdered manganese dioxide and Carbon.

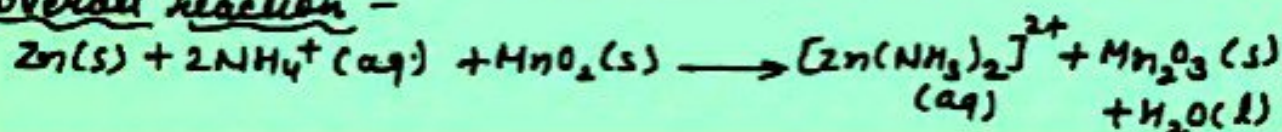
The space between the electrode is filled by a moist paste of NH_4Cl and ZnCl_2 .

The reactions are -





Overall reaction -



→ By using Nernst equation -

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{2.303 RT}{nF} \ln \frac{[\text{Zn}(\text{NH}_3)_2]^{2+}}{[\text{NH}_4^+]^2}$$

Due to the presence of ions in the overall reaction, its voltage decreases with time.

Ques: 4. Give Reasons -

- E° value for $\text{Mn}^{3+}/\text{Mn}^{2+}$ couple is much more positive than that for $\text{Fe}^{3+}/\text{Fe}^{2+}$
- Iron has higher enthalpy of atomization than that of Copper
- Sc^{3+} is colourless in aqueous solution whereas Ti^{3+} is coloured.

[All India 2018]

Ans - (a) $E_{\text{Mn}^{3+}/\text{Mn}^{2+}}^{\circ}$ = more positive

$E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^{\circ}$ = less positive

$\text{Mn}^{2+} = 3d^5 4s^0$

$\text{Mn}^{3+} = 3d^4$

→ Mn^{3+} will have a tendency to go in the +2 state because Mn^{2+} is more stable than Mn^{3+} , whereas in the case of Fe, Fe^{3+} is more stable than Fe^{2+} .

(b) Iron has more number of unpaired electrons. Hence, it will have stronger interatomic interaction than Copper.

(c) Sc^{3+} is colourless as it does not have unpaired electrons.
 Ti^{3+} has one unpaired electron. Hence, it is coloured.